# SAR-Based Flood Mapping

## Cyclone Gabrielle (New Zealand) and Musi River Flooding (India)

*A Machine Learning Approach Using Sentinel-1 Radar Imagery, with a Cross-Region Generalisation Test*

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# Executive Summary

This report presents a machine learning approach to flood extent mapping using Sentinel-1 Synthetic Aperture Radar (SAR) imagery. A Random Forest classifier was trained on Cyclone Gabrielle's flooding of Hawke's Bay, New Zealand (13-14 February 2023), using an official government-published flood extent layer as ground truth, and then applied without retraining to a separate flood event on the Musi River in Hyderabad, India (September 2024), to test whether the model generalises across a structurally different climate, terrain, and hemisphere.

The trained model achieved 97.4% test accuracy on the Hawke's Bay validation set. An ablation study isolating only the temporal change signal (excluding raw backscatter values) still achieved 79.7% accuracy alone, confirming a genuine flood-detection signal rather than memorisation of static terrain. When transferred to Hyderabad with no retraining, the model correctly identified the Musi River channel as flooded, but also produced explainable false positives over permanent water bodies and a likely airport runway - a known limitation of SAR-based water classifiers that is reported transparently in this study rather than concealed.

# Key Finding

**Training result:** A Random Forest classifier (200 trees, six features per pixel: pre-flood VV, pre-flood VH, post-flood VV, post-flood VH, and the VV/VH change between them) was trained on a balanced sample of 40,000 pixels drawn from the Hawke's Bay study area, using LINZ's official Cyclone Gabrielle flood extent layer as ground truth. Test accuracy on a held-out 30% split was 97.4%, with balanced precision and recall (0.97-0.98) on both the flooded and non-flooded classes.

**Ablation study:** To verify the model was learning genuine flood-change signal rather than simply memorising which areas are typically low-lying or wet, a second model was trained using only the two change features (VV and VH difference between pre- and post-flood images), with the raw backscatter values excluded entirely. This change-only model still achieved 79.7% accuracy on its own - well above chance, confirming a real temporal detection signal - but well below the full model's 97.4%, indicating that terrain and baseline wetness characteristics (captured by the raw pre-flood values) contribute substantially to the full model's performance. This is disclosed transparently: the model should be understood as combining genuine flood change detection with terrain susceptibility context, not as a pure change detector.

**Cross-region transfer test:** The trained Hawke's Bay model was applied directly to a Sentinel-1 pair covering the Musi River flooding in Hyderabad, India (pre-flood 25 August 2024, post-flood 6 September 2024), with no retraining or fine-tuning. The model correctly identified the Musi River channel as flooded. It also flagged two large areas consistent in location and shape with Osman Sagar and Himayat Sagar, two of Hyderabad's permanent reservoirs, and a geometric feature consistent with a runway near Rajiv Gandhi International Airport, which sits at the edge of the study area. These are very likely false positives: smooth surfaces such as permanent water and paved runways produce a similar specular radar reflection to flood water, and a model relying partly on raw backscatter values (as this one does, per the ablation study) cannot fully distinguish 'always wet' from 'newly flooded' without additional context such as a permanent water mask.

# Methodology

**Data source.** Sentinel-1 GRD imagery (IW mode, VV+VH polarisation), accessed via Google Earth Engine. Pre- and post-flood image pairs were selected by searching a date window around each flood event and matching the orbit pass (ascending/descending) between the pre- and post-flood images, to avoid introducing false backscatter differences purely from a change in viewing geometry.

**Study areas.** Hawke's Bay, New Zealand (pre-flood 9 Feb 2023, post-flood 14 Feb 2023, both ascending pass) and Hyderabad, India, covering the Musi River corridor and Old City (pre-flood 25 Aug 2024, post-flood 6 Sept 2024).

**Preprocessing.** A 5x5 median filter was applied to all SAR bands prior to feature extraction. SAR imagery has inherent per-pixel speckle noise from the coherent radar imaging process; without despeckling, pixel-by-pixel classification produced widespread scattered false positives with no relationship to actual flooding, visually confirmed before and after this step. Despeckling improved Hawke's Bay test accuracy from 92.6% to 97.4%.

**Reference data.** The Hawke's Bay Regional Council and LINZ jointly published an official, validated Cyclone Gabrielle flood extent polygon layer, compiled from aerial and satellite imagery, ground inspections, and hydrological modelling. This was rasterised to match the SAR grid and used as ground truth.

**Training.** A balanced sample of 20,000 flooded and 20,000 non-flooded pixels was drawn from the labelled Hawke's Bay raster and split 70/30 for training and testing. A Random Forest classifier (200 trees, max depth 15) was trained using scikit-learn.

# Results

## Hawke's Bay: Reference vs Model Prediction

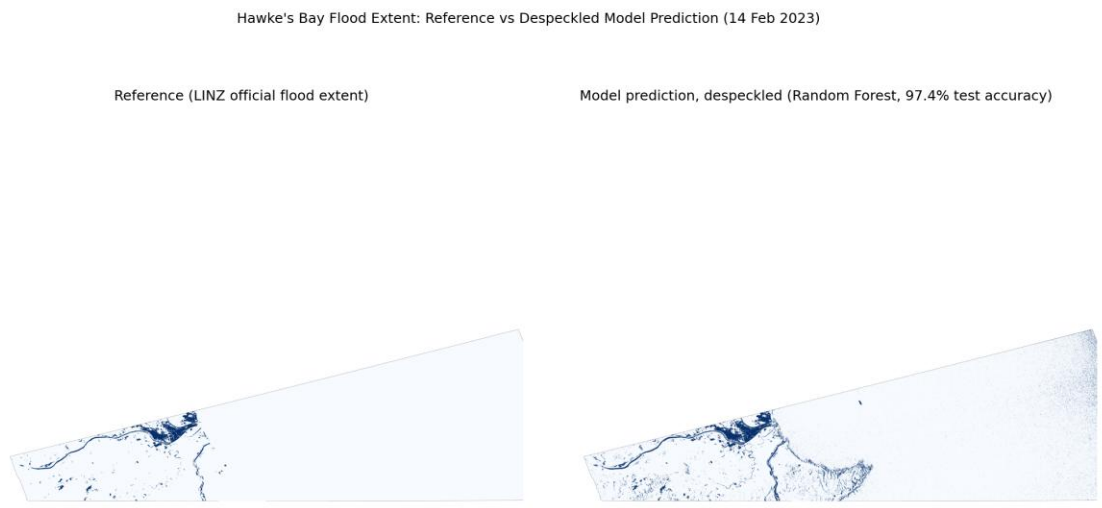


*Figure 1. LINZ official flood extent (left) vs the despeckled Random Forest model's prediction (right), 14 Feb 2023. The model closely reproduces the reference extent, including the river channel and the Napier-area flood cluster.*

## Cross-Region Transfer: Hyderabad

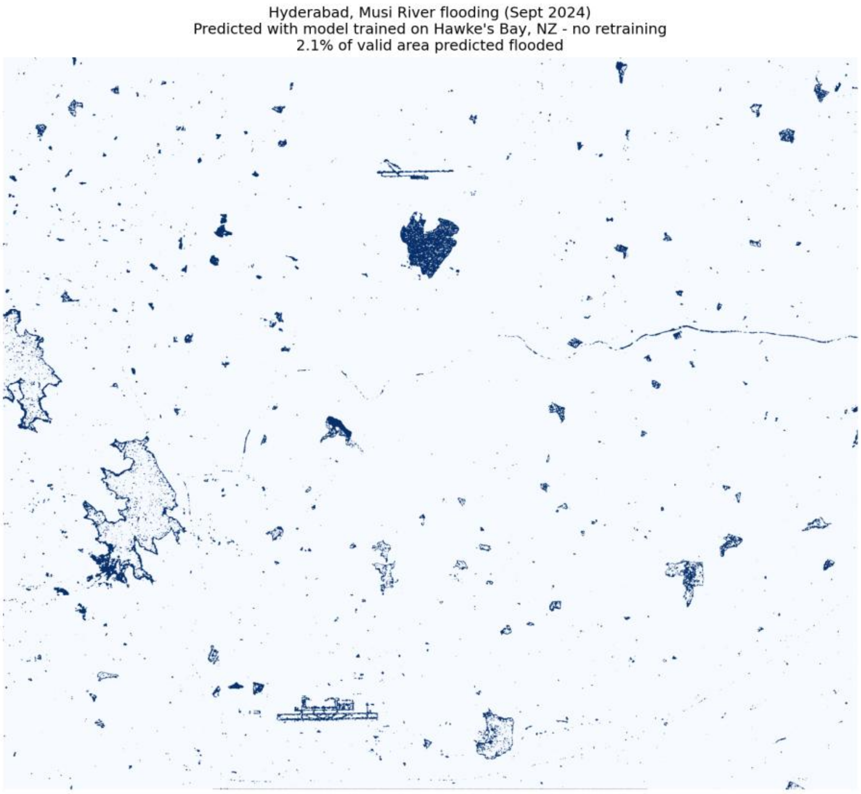


*Figure 2. The Hawke's Bay-trained model applied to Hyderabad's Musi River flooding (Sept 2024) with no retraining. The river channel is correctly identified; the two large irregular features on the left are likely permanent reservoirs (Osman Sagar, Himayat Sagar) and the geometric feature near the bottom is likely the Rajiv Gandhi International Airport runway - both explainable false positives from smooth-surface radar reflection.*

# Discussion

The results demonstrate both the potential and the limitations of a straightforward SAR backscatter classifier for flood mapping. Within the training region, accuracy against an independent, government-validated reference is excellent (97.4%). The ablation study shows this performance is not solely a product of memorising terrain, a genuine, if weaker, signal (79.7%) is present in the temporal change alone.

The cross-region transfer test is the more instructive result. The model generalises well enough to correctly trace a river channel it has never seen, in a different hemisphere and climate, without any retraining - a meaningful demonstration of transferability. But it also exposes a specific, well-understood weakness: any classifier that relies partly on absolute backscatter values (rather than change alone) will struggle to distinguish permanent water and other smooth, radar-reflective surfaces from genuinely new flooding, unless it is combined with an auxiliary permanent-water mask (such as the JRC Global Surface Water dataset) or restricted to change-based features at some accuracy cost. This is reported here as a finding, not concealed, since understanding a model's failure modes is as important as reporting its headline accuracy.

# Conclusion

A Random Forest classifier trained on Sentinel-1 SAR imagery and an official flood extent reference achieved strong, validated accuracy (97.4%) in its training region, with a documented ablation study confirming genuine change-detection capability. Applied without retraining to a structurally different flood event on another continent, the model correctly traced the flooded river channel while also revealing a known and explainable class of false positive (permanent water bodies, smooth artificial surfaces). This combination of a strong validated result, an honest ablation check, and a transparent account of cross-region limitations reflect the standard expected of applied geospatial machine learning work, rather than a single unqualified accuracy figure.

## Data Sources

- Sentinel-1 GRD (Copernicus), via Google Earth Engine - SAR backscatter (VV, VH), 10m resolution
- Cyclone Gabrielle Flood Areas (14 Feb 2023), LINZ Data Service - training/validation ground truth

## References

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## Repository

Full code, output maps, and trained model: github.com/Man3u/Flood-Mapping-SAR-NZ-India